

ECO387C - Fall 2024

Problem Set 10 Solutions

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Question 1 - Lucas Tree

1. Taking the pricing function $p(d)$ as given, the representative household's Bellman equation is the following:

$$v(a, d) = \max_{\{a'\}} \left\{ \frac{[(d_t + p_t)a_t - p_t a_{t+1}]^{1-\sigma} - 1}{1-\sigma} + \beta \sum_{d'} v(a', d') \pi(d'|d) \right\}$$

2. The recursive competitive equilibrium in this economy is the value function $v(a, d)$, asset pricing function $p(d)$, and asset purchasing policy function $g(a, d)$ such that:
- (a) Taking $p(d)$ as given, $v(a, d)$ and $g(a, d)$ solves the representative consumer's dynamic programming problem, and
 - (b) All markets clear in all periods
 - i. $c = d$ (goods market)
 - ii. $g(k, d) = 1$ (asset market)
3. To ease the notional burden, let $(a, d) \in X \subset \mathbb{R}^2$, and $B(X)$ be the set of bounded and continuous functions $V : X \rightarrow \mathbb{R}$.

Blackwell's sufficiency conditions require the Bellman operator T to satisfy the following two requirements:

- (a) **Discounting:** We need to show that $\exists \beta \in [0, 1)$ such that $T(v(a, d) + k) \leq T(v(a, d)) + \beta k$

$$\begin{aligned} T(v(a, d) + k) &= \max_{\{a'\}} \left\{ \frac{[(d_t + p_t)a_t - p_t a_{t+1}]^{1-\sigma} - 1}{1-\sigma} + \beta \left(\sum_{d'} v(a', d') \pi(d'|d) \right) \right\} \\ &= \max_{\{a'\}} \left\{ \frac{[(d_t + p_t)a_t - p_t a_{t+1}]^{1-\sigma} - 1}{1-\sigma} + \beta \sum_{d'} v(a', d') \pi(d'|d) + \beta k \right\} \\ &= T(v(a, d)) + \beta k \end{aligned}$$

(b) **Monotonicity:** Let $w(a, d) \geq v(a, d) \forall X$. We need to show that $T(w(a, d)) \geq T(v(a, d))$.

$$\begin{aligned} T(w(a, d)) &= \max_{\{a'\}} \left\{ \frac{[(d_t + p_t)a_t - p_t a_{t+1}]^{1-\sigma} - 1}{1-\sigma} + \beta \sum_{d'} w(a', d') \pi(d'|d) \right\} \\ &\geq \max_{\{a'\}} \left\{ \frac{[(d_t + p_t)a_t - p_t a_{t+1}]^{1-\sigma} - 1}{1-\sigma} + \beta \sum_{d'} v(a', d') \pi(d'|d) \right\} \\ &= T(v(a, d)) \end{aligned}$$

Hence, Blackwell's sufficiency conditions are satisfied, and T is a contraction mapping.

4. The representative consumer's first order condition with respect to a' is:

$$c^{-\sigma} p(d) = \beta \sum_{d'} v_1(a', d') \pi(d'|d)$$

Applying the Envelope condition gives us:

$$v_1(a, d) = c^{-\sigma}(p(d) + d)$$

Evaluating this at the optimum and substituting this back into the first order condition gives the representative consumer's consumption-Euler equation:

$$c^{-\sigma} p(d) = \beta \sum_{d'} (c')^{-\sigma} p(d') + d' \pi(d'|d)$$

Rearranging and substituting in the goods market clearing condition gives us:

$$p(d) = \sum_{d'} \left(\frac{d'}{d} \right)^{-\sigma} (p(d') + d') \pi(d'|d)$$

as required.

5. Now that the d' is I.I.D. over time, i.e. $\pi(d'|d) = \pi(d')$, the functional equation above simplifies to:

$$p(d) = \sum_{d'} \left(\frac{d'}{d} \right)^{-\sigma} (p(d') + d') \pi(d')$$

As stated in the problem set, make the guess that $\frac{p(d)}{d^\sigma} = \alpha$. Substituting this guess into the above gives:

$$\begin{aligned} \frac{p(d)}{d^\sigma} &= \beta \sum_{d'} \left(\frac{p(d')}{(d')^\sigma} + (d')^{1-\sigma} \right) \pi(d') \\ \alpha &= \beta \sum_{d'} (\alpha + (d')^{1-\sigma}) \pi(d') \\ &= \alpha\beta + \beta\delta, \end{aligned}$$

where

$$\delta \equiv \sum_{d'} (d')^{1-\sigma} \pi(d') = \mathbb{E}[d^{1-\sigma}]$$

Hence, $\alpha = \frac{\beta\delta}{1-\beta} = \frac{p(d)}{d^\sigma}$. Thus,

$$p(d) = \frac{\beta\delta}{1-\beta} d^\sigma, \quad \delta = \mathbb{E}[d^{1-\sigma}]$$

6. Substituting in the details of the 2 state-Markov chain, the prices for the Lucas tree now must satisfy the following:

$$p(d_L) = \beta \left[\frac{d_L}{d_H} [p(d_H) + d_H] \times (1 - p) + \frac{d_L}{d_L} [p(d_L) + d_L] \times p \right] \quad (1)$$

$$p(d_H) = \beta \left[\frac{d_H}{d_H} [p(d_H) + d_H] \times q + \frac{d_L}{d_L} [p(d_L) + d_L] \times (1 - q) \right] \quad (2)$$

This is now a system of two equations in two unknowns (p_L, p_H). Substituting the parameter values results in the following two equilibrium prices:

$$p(d_L) \approx 21.6$$

$$p(d_H) \approx 26.4$$